Department of Mathematics and Applied Mathematics

3MS Metric Spaces

Tutorial 9 **2024**

1. You may recall from real analysis that, in $(\mathbb{R}, d_E)$, all Cauchy sequences are bounded. This result extends to arbitrary metric spaces, in the form given by the lemma below.

   Lemma Let (x_n) be a Cauchy sequence in a metric space (X, d).
   Then the diameter of the set $\{x_n : n \in \mathbb{N}\}$ is finite.

   Proof Choose $N \in \mathbb{N}$ such that $m \geq n \geq N$ implies that $d(x_n, x_m) < 1$. (1)
   Let $M = \max\{d(x_r, x_s) : r = 1, \ldots, N \text{ and } s = 1, \ldots, N\}$. (2)
   Then $d(x_p, x_q) \leq M + 1$ for all $p, q \in \mathbb{N}$. (3)
   So diam $\{x_n : n \in \mathbb{N}\} \leq M + 1$. (4)

   (a) Why do we know that an N as mentioned in line (1) exists?

   (b) In line (2), we take the maximum of a set. How do we know that maximum exists?

   (c) Prove the claim in line (3).

   (d) Why does line (4) follow from line (3)?

2. (a) The argument below attempts to prove that every subspace of a complete metric space is complete. Provide a counter-example to show that this statement is false.

   (b) What is the mistake in this argument? (If it is not clear, you may want to work through the argument with your example above, and see what happens.)

   Suppose that X is a complete metric space and B is a subset of X. (1)
   We use Cantor's Intersection Theorem to show that B is complete. (2)
   Let $A_1 \supseteq A_2 \supseteq A_3 \supseteq \ldots$ be a decreasing sequence of non-empty subsets of B with diam $A_n \to 0$. (3)
   We must show that $\bigcap_{n=1}^{\infty} \overline{A}_n \neq \emptyset$. (4)
   Since $A_n \subseteq B$ for all n and $B \subseteq X$, the A_n's form a decreasing sequence of non-empty subsets of X with diam $A_n \to 0$, (5)
   and so by completeness of X, $\bigcap_{n=1}^{\infty} \overline{A}_n \neq \emptyset$ as required. (6)

3. We call a metric space X **absolutely closed** if it satisfies this condition: for every metric space Y, if X is a subspace of Y, then X is a closed subset of Y.
   Prove that a metric space is complete if and only if it is absolutely closed.

4. Consider $(\mathbb{R}, d_E)$ and let $f : \mathbb{R} \to \mathbb{R}$ be a function defined by $f(x) = \frac{1}{7}(x^3 + x^2 + 1)$.

   (a) Is f a contraction on $\mathbb{R}$?

   (b) Show that f maps $[0, 1]$ into $[0, 1]$.

   (c) Is f a contraction on $[0, 1]$?

   (d) How many fixed points does f have in $[0, 1]$?

   (e) If you use $x_1 = 0$ as a starting point and use iteration to try and find a fixed point for f in $[0, 1]$, estimate the error in your approximation to the fixed point after n steps. [Do not try to find x_n].

   (f) How many fixed points does f have in $\mathbb{R}$? Is it possible to find them all using the contraction mapping theorem? Give reasons to your answers.

5. Consider the function $g(x) = e^{-x} + 2$ on the domain $([1, \infty), d_E)$.
   Show that Banach's Fixed Point Theorem can be applied to this function: check the hypotheses carefully, and then write down what the theorem tells you.
   You may want to remember the Mean Value Theorem, which says that for a differentiable function f on an interval $[a, b]$, there exists c between a and b such that $f(b) - f(a) = f'(c)(b - a)$.
   If you feel like it, start at $x_1 = 1$, say, and look at a few terms of the appropriate sequence to see that it does what you expect.